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Two-Layered and Stretchable e-Textile Patches for Wearable Healthcare Electronics

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Wearable healthcare systems require skin-adhering electrodes that allow maximal comfort for patients as well as an electronics system to enable signal processing and transmittance. Textile-based electronics, known as "e-textiles," is a platform technology that allows comfort for patients. Here, two-layered e-textile patches are designed by controlled permeation of Ag-particle/fluoropolymer composite ink into a porous textile. The permeated ink forms a cladding onto the nanofibers in the textile substrate, which is beneficial for mechanical and electrical properties of the e-textile. The printed e-textile features conductivity of ≈3200 S cm^{-1}, whereas 1000 cycles of 30% uniaxial stretching causes the resistance to increase only by a factor of ≈5, which is acceptable in many applications. Controlling over the penetration depth enables a two-layer design of the e-textile, where the sensing electrodes and the conducting traces are printed in the opposite sides of the substrate. The formation of vertical interconnected access is remarkably simple as an injection from a syringe. With the custom-developed electronic circuits, a surface electromyography system with wireless data transmission is demonstrated. Furthermore, the dry e-textile patch collects electroencephalography with comparable signal quality to commercial gel electrodes. It is anticipated that the two-layered e-textiles will be effective in healthcare and sports applications.

1. Introduction

Textiles are inherently "smart materials" with several unique characteristics that have been optimized for human comfort over thousands of years, such as softness, thermal dissipation, breathability, light weight, and high conformability. Recently, embedding electronics in textiles, termed "e-textiles," has been suggested to be a next-generation technology. For personalized healthcare devices, ultimate user comfort may be achieved with e-textiles.[1,2] In recent years, the use of e-textiles has shown tangible outcomes with the aid of advanced nanotechnologies.[3] A complete form of fully integrated e-textile systems include semiconducting transistors,[4] processing units,[5] sensors,[6] generators,[7] and energy storage units.[8]

To achieve a fully integrated e-textile, stretchable interconnects (sometimes referred as conducting traces) are central to the seamless integration of functional devices with the desirable mechanical properties of the substrate textiles. In the last decade, such conducting traces in smart e-textiles have been realized by: i) interweaving metallic or polymeric 1D thread conductors into textiles,[9] ii) fully coating 2D textile sheets with conductive materials,[10] or iii) patterning or printing the traces on 2D textile/fabric substrates.[3e,7a,11] Strategy (i) has been challenged by the limited material selection of the electronic threads. Strategy (ii) suffers from the lack of deterministic pattern ability, thus miniaturization is not a straightforward task. Strategy (iii) is currently prevailing in a large spectrum of applications with the aid of recent advances in materials and printing techniques. Recently, Someya and co-workers achieved screen printable viscous composite inks with a low surface resistivity of 0.06 Ω sq^{-1} by processing a multilayered print (five overlays) under pressure (30 kPa) and at high temperature (160 °C); this technique produced an e-textile capable of mechanically withstanding cyclic stretching (1000 cycles at 10% maximum strain) with a resulting order of magnitude increase in electrical resistance.[11b,12] A drawback of the process may be the involvement of hot-press processing, which may cause a thermally induced degradation of textile properties as most commodity textiles degrade at temperatures between 125 and 180 °C.[13]

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Stretchable electronics with physiological sensing abilities, such as electrical signals,[14] electrochemical signals from sweat,[15] temperature,[16] tactile force,[17] blood glucose,[18] blood oxygenation, and pulse,[19] have been a new leading trend in wearable healthcare. Meanwhile, e-textiles promise an intimate and comfortable integration to the human body. The necessary first step toward the application of e-textiles in healthcare is development of the sensory patches that are in direct contact with human skin.[20] For meaningful healthcare applications, such e-textile sensory patches connect with electrical circuits that enable signal transduction and wireless communication abilities.[21]

In this work, we develop a stretchable, yet mechanically and electrically robust e-textile sensory patch along with an electronic system for wearable healthcare applications. Here, silver-based fluoroelastomer composite ink forms conducting traces via jet printing, where the ink's penetration depth into electrospun porous polyurethane textile is controlled by utilizing the size and shape of silver particles, thus enabling two-layer configuration (**Figure 1**). Compared to the conventional single-layer structure for sensory patches, the two-layer structure enables a simple and reliable definition of electrode areas while the conducting traces are intrinsically separated from human skin. The traces in the e-textile show a high conductivity of ≈3200 S cm^{-1} with high stretchability and durability over repeated strain cycles. Such high conductivity and durability are achieved by a one-step printing process which does not require any post-treatment. Finally, we demonstrate the biomedical

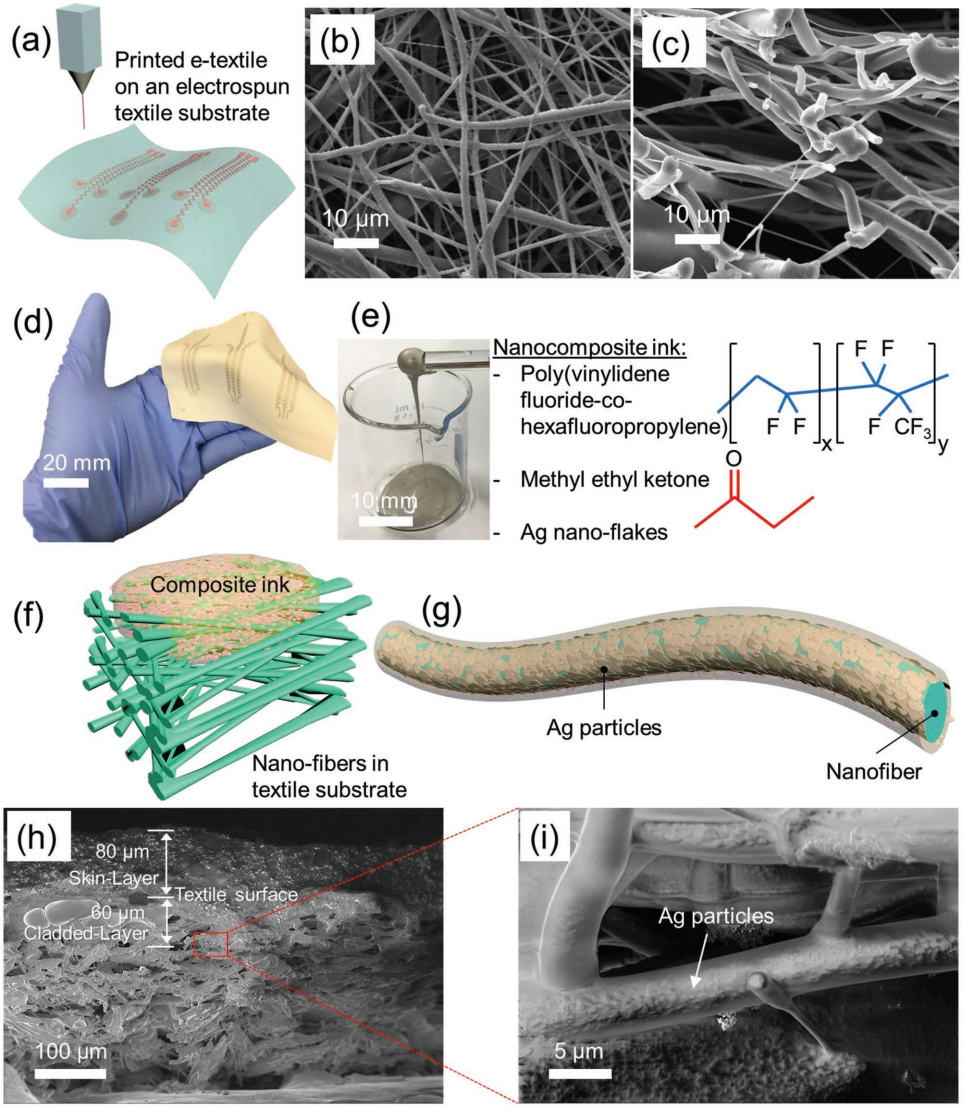


Figure 1. Printed, stretchable e-textile conductors. a) The conceptual drawing of jet-printing nanocomposite inks onto a textile substrate. Scanning electron microscope (SEM) image of the textile substrate, which was made of compressed electrospun nanofibers of polyurethane: b) top view and c) cross-section view. d) A sheet of e-textile with printed conductive serpentine traces. e) Prepared nanocomposite ink which consisted of silver particles, fluoroelastomer, and methyl ethyl ketone. f) A schematic illustration of printed ink on the textile substrate. g) A schematic illustration of a single cladded nanofiber in the cladded-layer of printed e-textile. SEM images of h) cross-sectional view of the e-textile after printing of the Ag-powder-based ink and i) the close-up view of the cladded layer.

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application of our e-textile-based systems for i) surface electromyography (sEMG) sensing of human muscles activities in a rehabilitation medicine setting and ii) electrocardiography (electroencephalography, EEG) sensing of human brain waves.

2. Results and Discussion

2.1. Printing Conductive Traces

The key advantage of our printing technology is that it is one-step processing without high-temperature post-treatment (see Figure 1a and Figure S1, Supporting Information). In contrast, available commercial silver ink requires thermal annealing at temperatures higher than 130 °C.[22] We printed a nanocomposite ink on a nonwoven thermoplastic polyurethane (PU) substrate (SNS Nanosan-Sorb[23]), in which the electrospun fibrous structure provides stretchability. The substrate is developed for human skin contact applications, thus is breathable for air permeation and slightly hydrophilic to absorb a moderate amount of sweat and contaminants. A scanning electron microscope (SEM) image of the textile substrate in Figure 1b,c shows fibrillar structure with highly polydisperse diameters ranging from 100 nm to 10 μm, which also provide an open porous structure for conductive ink infusion. The 300 μm thick textile is reported to have a stretchability up to 250% when uniaxially strained, whereas full dimensional recovery is available below 50% of strain. The soft and conformal nature of the e-textile is preserved after the printing process (Figure 1d).

The composite ink was prepared by mixing conductive silver particles (Sigma-Aldrich, and two species of the conductive particles were used as silver flakes with average particle size of 10 μm,[24] and silver powders with average particle size of 2–3.5 μm[25]) and flouroelastomer (DAI-EL G-801, DAIKIN) in methyl ethyl ketone (MEK) (Figure 1e). The prepared viscous ink was loaded to a jet-printer (Tabletop-3Dn, nScrypt). Ink droplets were expelled via a 150 μm nozzle at a pressure of 10 kPa relative to atmospheric pressure (controlled by the software associated with the printer) onto the textile substrate, which was oxygen plasma treated right before the printing process. Upon deposition, the textile substrate absorbed a certain amount of the ink microdroplets into its porous internal structure. When the solvent in the microdroplet dried in a fume hood at 22 °C for 8 h, an 80 μm thick skin-layer of dried ink forms on the surface of the textile substrate. At the same time, there formed a 60 μm deep cladded-layer, wherein the conductive ink forms cladding along the nanofibers in the inner part of the textile substrate (Figure 1f–i). This cladded-layer, which is thermodynamically stable as a coating layer (see the Supporting Information), increases the mechanical durability, conductivity, and adhesion of conductors to the textiles.[26] These advantages play key roles in the performance of e-textiles when integrated to a full electronic system. In addition, after the solvent is dried out, the conductor is typically consisting of Ag particles and flouroelastomer both of that are nontoxic for skin contacts, whereas the commercial polyurethane substrate is inherently designed as a decontaminant for skin; however, further tests are necessary to establish our e-textile's biocompatibility, especially for prolonged use.

2.2. Effects of Ag Particles on Permeation of Composite Ink into Nanotextiles

In this work, we used two species of Ag particles with different shapes and average particle sizes, including: i) flakes with the *average size* of 10 μm metal traces (denoted as Ag flakes hereafter) and ii) powders with *average size* of 2–3.5 μm metal traces (denoted as Ag powders). The "*average size*" values are stated as we received from the vendor (Sigma-Aldrich); however, the actual size distribution of individual particles (flakes or powders) was extremely polydisperse in nature. Especially, the existence of nanosized particles seems to affect the properties of e-textiles. The SEM image of the Ag flakes reveal that each particle is predominantly larger than a micrometer in size with irregular shapes (**Figure 2**a), whereas many particles in the Ag powders had sizes around (or even less than) 200 nm (Figure 2b). These small particles efficiently permeate into the pores of the textile substrate, resulting in the formation of cladded-layer shown in Figure 1f,h. Figure 2c,d shows the cross-section of the *printed inks* (i.e., Ag particles *with fluoropolymers* after solvents were dried). The printed ink containing Ag flakes showed distinctively smaller thickness of cladded-layer (Figure 2c) when compared to the printed ink with Ag powders (Figure 2d).

The thickness of the cladded-layer, which is equivalent to the permeation depth of liquid ink into the porous textile substrate, is the result of interplay between ink viscosity, ink-jetting parameters, solvent drying rate, and wetting. In order to study the effect of wetting, we prepared a smooth and nonporous version of the textile substrate by pressing with mild heating at 80 °C (see details in the Supporting Information). The measured contact angle values of the *wet* inks (i.e., *with* the solvent) were 65.16° and 40.20° for Ag flake and Ag-powder-based inks, respectively (Figure S4, Supporting Information). Both values were below 90°, which implies that both inks would intrinsically prefer to permeate into the textile substrate, whereas the wet Ag-powder-based inks have a higher tendency to penetrate into pores, owing to their lower contact angle. In addition, the permeation depth and absorbed ink amount were kinetically limited because the process competes with solvent drying.[12,26] In our experiment, viscosity, ink-jetting parameters, and solvent drying rate were essentially the same for the two types of inks, but the permeation depth showed a stark difference. Additionally, when composite materials are printed into a porous substrate, one must take the size effect of printing particles and the pores in the substrate into account. As shown in Figure 2c,d, Ag-powder-based ink contains many small-sized particles that can freely pass through the pores opening through the forest of nanosized fibers in the substrate, whereas Ag-flake-based ink does not. As a result, the Ag-powder-based ink was able to form a cladded layer of around 60 μm in thickness (Figures 1h and 2d and Figure S6, Supporting Information), whereas that of Ag-flake-based ink was only around 10 μm thick (Figure 2c). We believe that such geometric effect dictated the difference in the permeation depth of the wet ink, which finally caused the thickness of the cladded-layer.

2.3. Electrical and Mechanical Properties of the Printed e-Textile

Mechanical and electrical properties of the trace-printed e-textiles were measured from a rectangular strip (**Figure 3**a) and a

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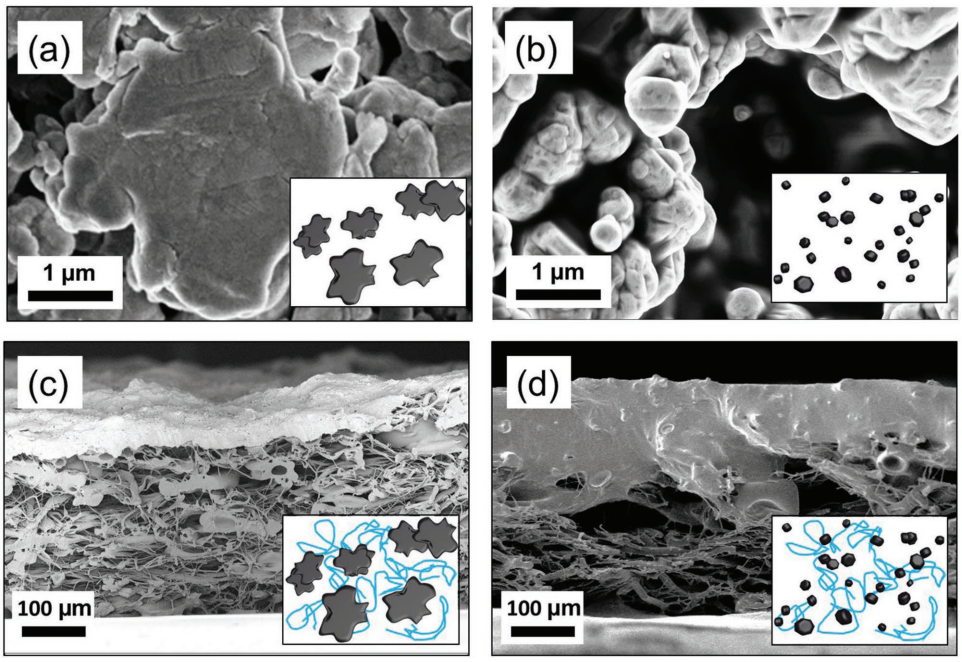


Figure 2. SEM images of a) Ag flakes b) Ag powders. Ag powders contain finer Ag particles than Ag flakes, as illustrated in the cartoons in the insets in (a) and (b). Cross-sectional SEM images printed e-textiles with c) Ag-flake-based and d) Ag-powder-based composite inks. The insets in (c) and (d) depict the existence of fluoropolymers in the composite inks (drawn not to scale).

serpentine trace (Figure 3b) printed with the composite inks. The rectangular strip was 4 mm in width and 40 mm in length. The serpentine traces had 0.5 mm linewidth, 2 mm lateral amplitude, 40 mm length, and 2 mm pitch interval (Figure S3, Supporting Information). After drying, the original values of resistance were 0.22 Ω for the rectangular strip and 4.1 Ω for the serpentine trace. Figure 3c shows the change in resistance ratio (the resistance value under strain normalized by the value at rest) under uniaxial strains from zero to 125%. The resistance of the e-textile sample with the rectangular strip increased 412-fold at 56% strain, whereas the increase was only 24.7 times of the original value at even higher strain of 120% in the serpentine sample.

Uniaxial stress–strain curves of the pristine textile and the two types of printed e-textiles were measured over five stretch-release cycles up to 50% strain (Figure 3d). As expected for viscoelastic materials, hystereses were observed in all measurements to different degrees. However, all samples showed full recovery (i.e., closed-loop cycles that end at zero strain when stress reduces to zero) at the end of each cycle. In addition, the uniaxial strain tests of the pristine and the printed textiles were performed (Figure S5, Supporting Information). The maximum strains at break were 115% for the pristine textile, 240% (Ag-flake-based ink) and 250% (Ag-powder-based ink) for the rectangular strip printed samples, and 120% (Ag-flake-based ink) and 122% (Ag-powder-based ink) for the serpentine printed samples. Detailed investigations on the increased stretchability and stiffness of the strip printed samples are currently underway.[26] All of the pristine and the printed textiles exhibited plastic deformation before rupture; similar behavior has been previously reported as characteristics of electrospun polyurethane fibers.[27]

Figure 3e shows cyclic mechanical reliability on electrical resistivity of the e-textiles with serpentine traces over a thousand stretch-release cycles. Here, three maximum strains, up to 10%, 20%, and 30%, respectively, were applied with a stretch/release strain rate of 4% per second. Although the resistance values gradually increased with increasing cycles, the relative increase in resistance values were within the same order of magnitude even after one thousand cycles even at 30% maximum strain.

2.4. Two-Layered e-Textile Patches

When the cladded-layer thickness (i.e., ink permeation depth) is controlled to be approximately a quarter of the textile thickness, as shown in Figure 1g, electrical insulation between the two sides of e-textile can be assured, whereas the cladded-layer helps in achieving better conduction under high strains, as shown in Figure 3c. Using this design concept, we fabricated a two-layered e-textile patch solely by ink printing, as shown in **Figure 4**a,b. Here, the sensory electrodes were printed on one side and the serpentine traces printed on the opposite side of the textile substrate. The separation of the electrodes and the traces allowed precise targeting of the sensory area on human skin as well as robust electrical insulation required for traces. In addition, an elastomeric thin-film (e.g., 200 μm thick polydimethylsiloxane (PDMS) Sylgard 184, or 250 μm thick 3M very high bond (VHB) double-side tape) was used to protectively encapsulate the serpentine traces and hinder detachment of Ag particles. The encapsulating thin-film was patterned by a scan-and-cut machine (Brother CM550DX) to fit the trace layouts. The VIA (vertical interconnect access) between the two layers

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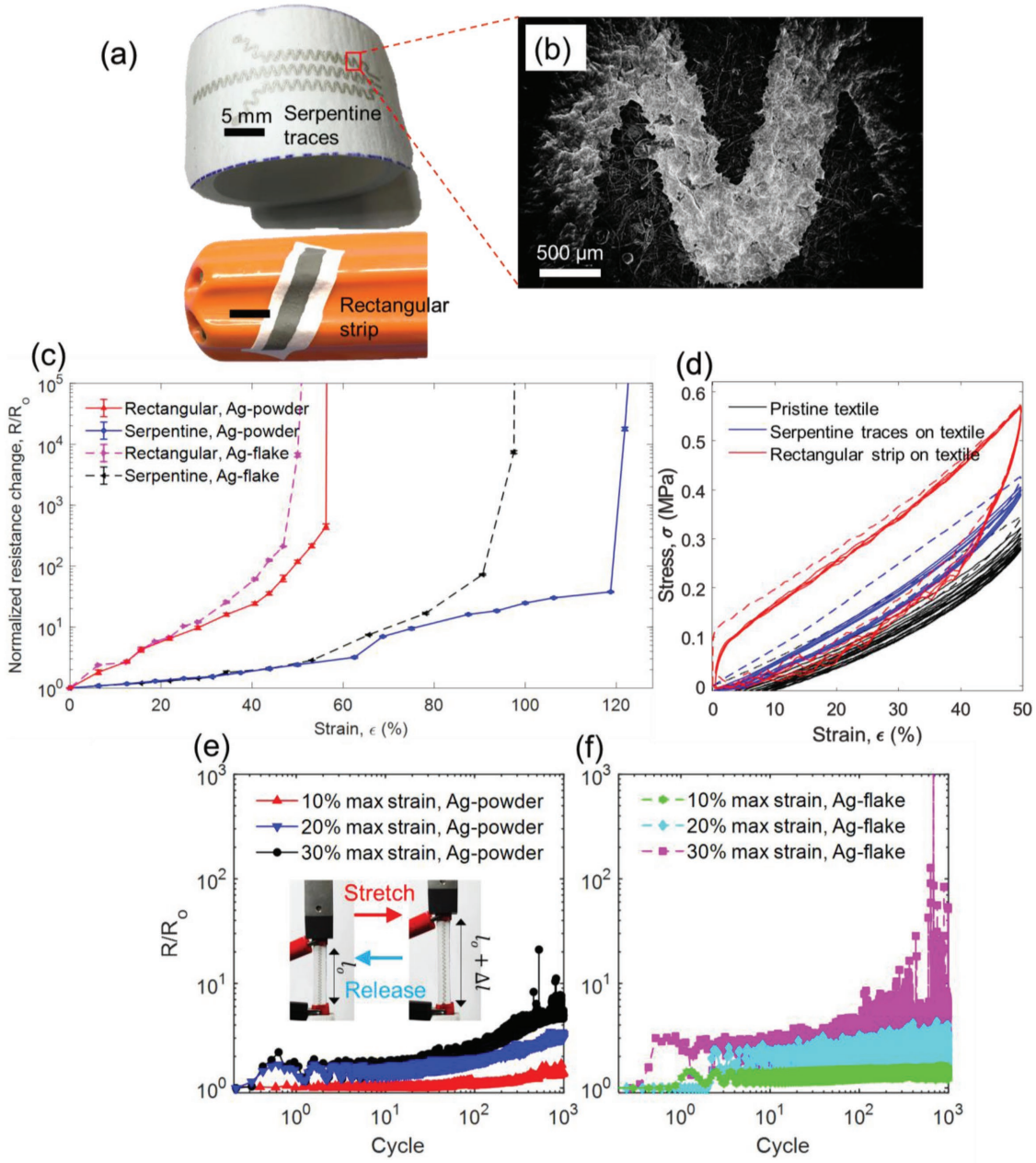


Figure 3. a) Stretchable printed e-textiles with a serpentine (up) and a straight (rectangular) shaped traces (as the e-textile was placed on a hand holder of a screwdriver). b) SEM image of the serpentine trace. c) Changes in resistance ratio of the serpentine and straight traces under uniaxial tensile stress conditions (as each data line represented the mean and standard deviation of five samples). d) Cyclic stress–strain curves (five cycles; dashed lines denote the first loop) for the pristine textile and printed e-textiles. Reliability tests from serpentine-printed e-textile samples wherein the composite inks contain e) Ag-powder and f) Ag-flake, respectively.

was formed simply by injecting a small amount of the ink by a needle (0.3 mm diameter) to allow complete permeation across the whole substrate thickness, as shown in the SEM images in Figure 4c,d. Figure 4e,f shows that the two-layered e-textile patches can withstand harsh stretching conditions. When applying on human skin, a medical grade double side tape (3M 1509 Medical Tape, thickness 0.12 mm) with relief patterns was used to achieve conformal attachment while maintaining air/moisture breathability (Figure S8, Supporting Information). When attached, the electrode-skin impedance was in the range of 120–150 kΩ with measured frequency of 10 Hz.

2.5. Fully Integrated E-Textile System for sEMG Application

The two-layered e-textile patch was connected to an integrated electronic system that comprises a signal amplifier, a noise-reducing filter, an analogue to digital converter, and a Bluetooth

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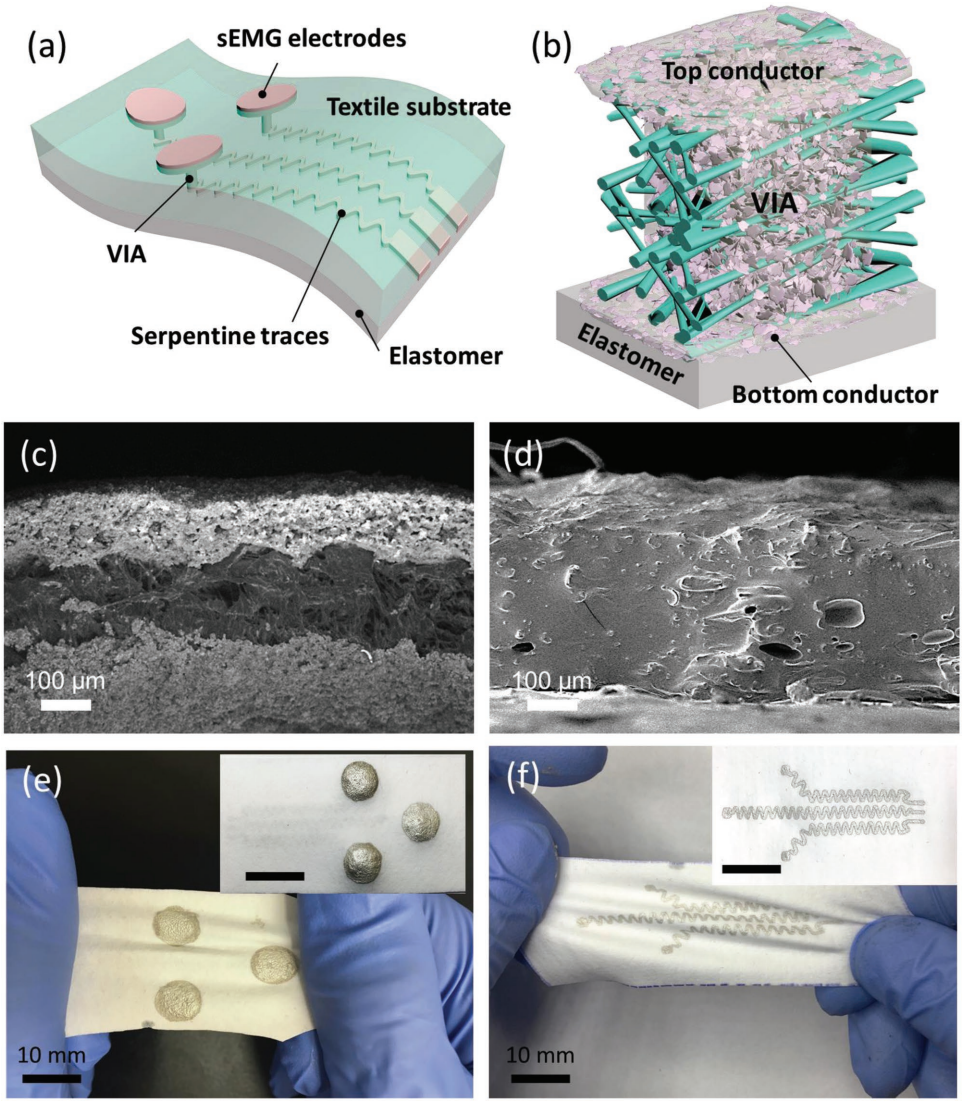


Figure 4. Two-layered printed e-textile patches. a) Schematic of the e-textile patch comprising three printed electrodes, VIAs, and serpentine traces encapsulated by dielectric elastomer (PDMS, or acrylics). b) An illustration of the VIA. Cross-sectional SEM images of printed two-layered e-textile where each of the layers are electrically c) insulated by the gap between cladded-layers and d) conducting through the dried-ink-filled VIA between the printed layers. Top view photographs of the two-layered e-textile seen from e) the sensory electrode and f) the serpentine traces sides. The main panels and the insets (scale: 10 mm) show stretched and nonstretched states, respectively.

wireless transmitter to export sEMG signals to data receiving electronics (**Figure 5**a). To attain a stable electrical connection between the e-textile patch and the circuit board, the e-textile patch was precisely aligned to connection pads on the circuit board, then clamped to the board (Figure S7, Supporting Information). The sEMG system is fully portable, lightweight, and compact (Figure 5b). As sensing electrodes for sEMG, the two layered e-textile has an excellent merit of eliminating signal contamination because the location of serpentine traces is intrinsically insulated from any skin-induced noises. In addition, the thin and conformal form factor of the e-textile further helps the suppression of unwanted artifacts from body movements, which are not associated with the electrical signals from the muscles of interest. Figure 5c demonstrates the block diagram of the integrated electronic system. Starting from the raw bipolar sEMG signal, the electronic system, amplified, filtered, biased, and digitized the signal at 1000 Hz, which was routed to a Bluetooth radio where it was transmitted in real-time to a computer for data analysis.

The signals from the sEMG patches were amplified with analog circuit to finely stay in the input voltage of the analog-to-digital (A/D) converter. Then, the microcontroller relayed the converted signal to the Bluetooth transceiver. The wireless signal can be sent to a computer or mobile device for further analysis or rehabilitation purposes. Here, the sEMG signal was digitally processed to measure the actual muscle activity versus noise. Muscle activity measured by EMG signal is generally in the frequency range of 1–500 Hz, with the main frequency components concentrated in 60–150 Hz.[28] Several factors such as muscle type, muscle activity, electrode-skin contact, electrode

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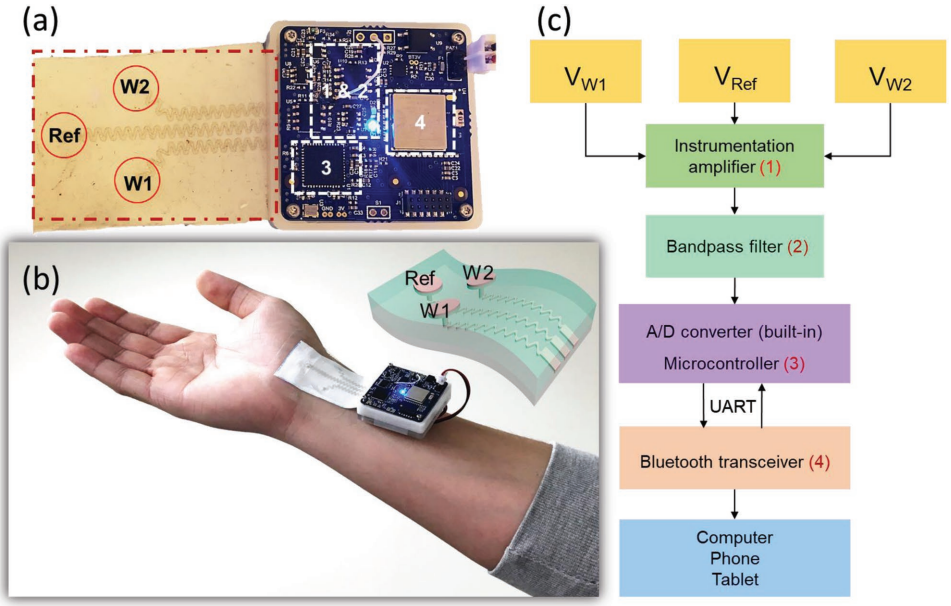


Figure 5. a) Fully integrated sEMG sensor system with the two-layered e-textile sensing electrodes for real-time monitoring muscle activities. b) Its placement on wrist muscle. c) Block diagram of the electronics system.

placements, fatigue, and task duration affect timing and frequency characteristics of the sEMG signal.

2.6. sEMG Monitoring of Muscle Activities

The sEMG system was worn on various body areas of a participant to monitor muscle activity from different parts of body, including i) the submental space, ii) elbow, iii) calf, and iv) ankle (**Figure 6**a). For each recording location, the reference electrode was located on or near a boney region while the remaining two electrodes were placed on the muscle belly of interest.[29] When necessary, the system was secured with medical tape to the skin to prevent the detachment of the e-textile due to the weight of the system.

To place the sEMG system on the submental space, the reference electrode was placed on the body of the mandible while the other two electrodes were over the suprahyoid muscles separately during saliva swallowing and mouth opening/closing.[29] The participant was asked to look straight ahead and swallow their saliva (five trials). Thereafter, they were instructed to open and close their mouth (five trials). While measuring the swallowing movements, positive and negative spikes occurred in the acquired signal (see in Figure 6b,c). Figure 6d,f shows the signal during mouth opening and closing events. While the peak signals occurred in both activities, the sEMG patterns from the swallowing events were followed by a longer duration of signal undershoot (about 2–4 s) after the initial peak. In order to record the hand crunching events, the sEMG system was located on the elbow of the participant. The participant was asked to keep the hand closed and then crunch/relax. As shown in Figure 6e,g, positive and negative peaks occurred during hand crunching and relaxing events, respectively. Similarly, the system attached on calf detected the signals from toe lifting (Figure 6h,j) and ankle bending (Figure 6I,k). In our work, an moving average filter and a band-pass filter (20–200 Hz) were used to process the raw sEMG signals, then median frequency between 1 and 500 Hz is calculated. We observed that the median frequencies of the sEMG signals were in the range of 60–150 Hz (see Figure 6b–k), which are in the frequency range of muscles activities as reported.[28] It indicates that the measured sEMG signal were representative of muscle electrical activities.

2.7. Monitoring Human Brain Activities

We further used the two-layered e-textile patches for EEG assessment of brain waves. **Figure 7**a illustrates the setup of EEG measurement, in which the EEG sensor (red color) was placed in the hairless area behind the ear (mastoid process), and another commercial gel-electrode was on the forehead area as a reference. For comparison, we simultaneously measured the EEG signals (red sensor) by i) the e-textile sensors (right mastoid) and ii) commercial rigid sensors (left mastoid). While conformability and stable contact are needed for high-quality recording, the EEG signals by the e-textile sensor showed fewer smaller movement artifacts than the commercial one, as can be observed by comparing the time (Figure 7b) and frequency domain (Figure 7c,d) outputs between the two types of electrodes. In the frequency domain spectrograms, it is apparent that there is some low-frequency (<10 Hz) noise in the flexible signal as well.

To test if the electrodes were measuring true brain signals, we used a classic paradigm to isolate increases in alpha oscillations (7–14 Hz) when individuals close their eyes.[30] The participant opened their eyes and stared at a fixation cross on a computer monitor in a sound and electrically isolated recording chamber. After 3 min, the participant closed their eyes for 7 min, then opened them and fixated for an additional 2 min. Figure 7e,f shows the frequency domain EEG spectra, averaged

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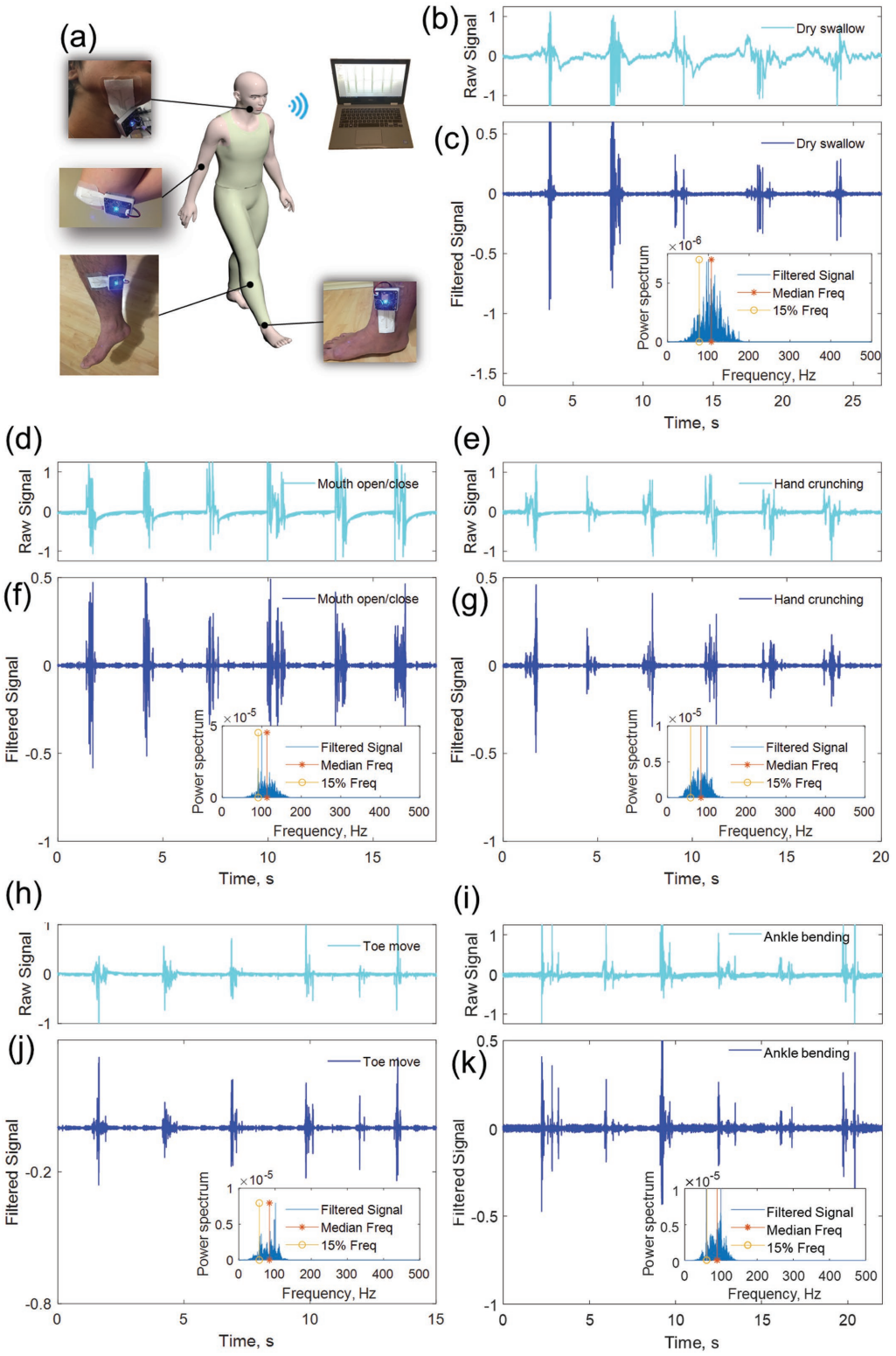


Figure 6. sEMG measurements of muscles activities. a) Schematic illustration of sEMG measurements from various parts of the human body. b) The raw and c) filtered sEMG signals (by using a moving average filter and a bandpass filter of 20–200 Hz) from dry swallowing episodes measured from the submental space. d) The raw and f) filtered sEMG signals from mouth open/close episodes measured from the submental space. e) The raw and g) filtered sEMG signals from hand crunching episodes measured from elbow. h) The raw and j) filtered sEMG signals from toe lifting episodes measured from the ankle. i) The raw and k) filtered sEMG signals from ankle bending episodes measured from the ankle. Here, all insets are power spectrums obtained by Fourier transforms of each data.

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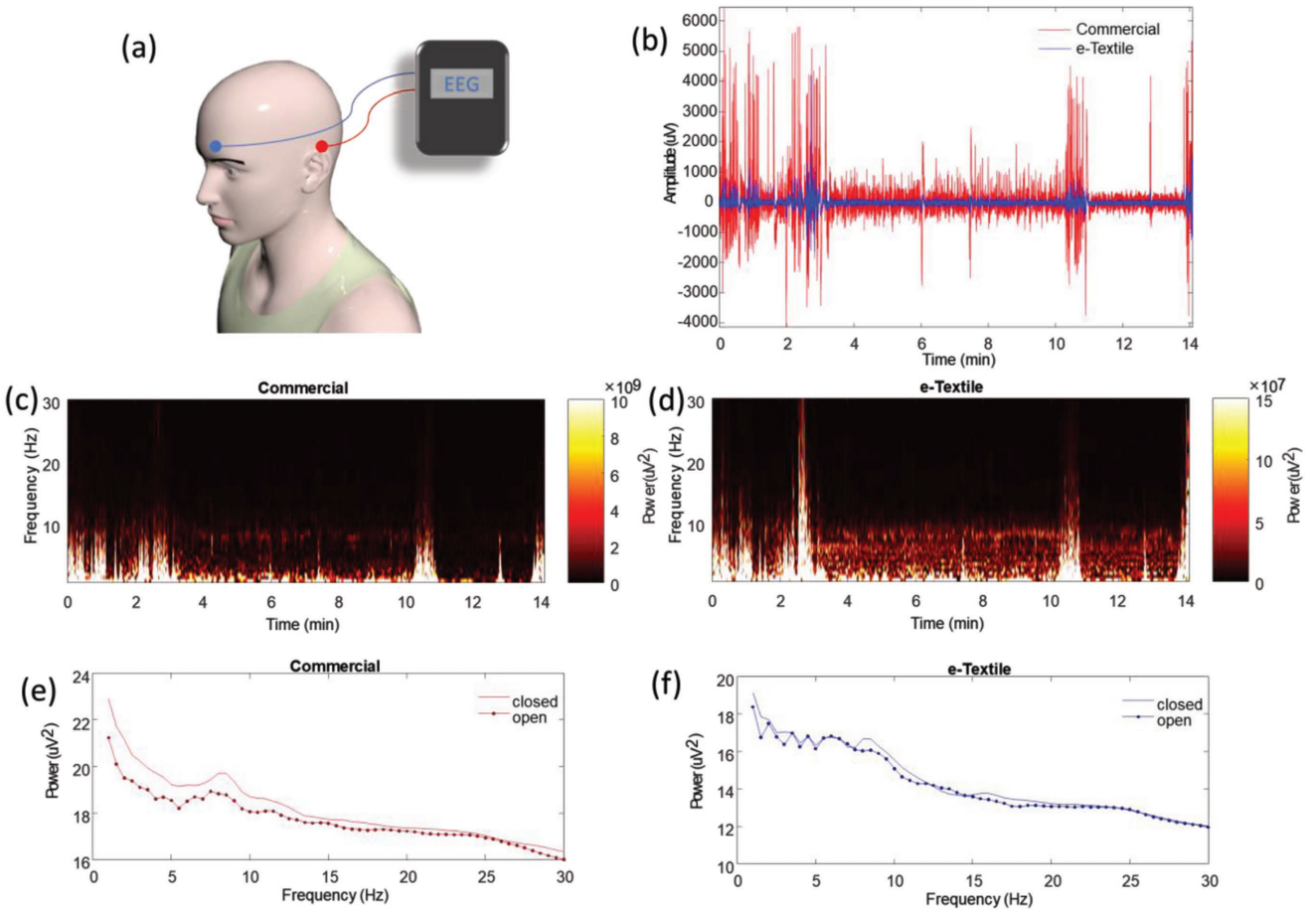


Figure 7. EEG measurements for human neuroscience. a) Schematic illustration of EEG recording from behind the ear. b) Raw data collected from a commercial gel EEG electrode behind the left ear, simultaneously with an e-textile electrode behind the right ear (blue). Larger spikes in noise are observed for the commercial sensor due to less mechanical conformation with the skin. c) Spectrogram computed with a wavelet analysis of the raw data, showing the frequency over time. At three minutes, the eyes are closed, and then again at 10 min they are opened. Evident are large artifacts due to muscle activity (0–10 Hz) as well as a band of alpha oscillations (≈9 Hz) indicative of human brain activity. d) The flexible e-textile electrode also measured these same signals as well as a slight increase in <10 Hz noise. e) and f) show the spectra, averaging over time, for the two electrode types. Evident is the typical $1/f$ shape indicative of human EEG as well as a clear peak around 9 Hz, the stereotypical alpha oscillations, that as expected are larger when the eyes are closed.

over time for each condition (eyes open, eyes closed). We found a clear $1/f$ spectrum in the EEG, stereotypical of human brain electrical activity, and also the expected difference (increased 7–14 Hz activity of alpha waves) between when the participant's eyes were opened (i.e., increased alpha) and when they were closed (i.e., decreased alpha), for both the e-textile and the commercial systems (also apparent in the spectrograms in Figure 7c,d as a horizontal band at ≈9 Hz). These results show great promise for future mobile EEG experiments and research.

3. Conclusion

In this work, two-layered e-textile patches with high mechanical durability and electrical performance were fabricated by jet-printing Ag-powder/fluoropolymer-based nanocomposite ink on both sides of the porous textile substrate. The interplay between wetting and particle size effect controlled the thickness of the cladded-layer, where the dried inks coat the surfaces of the nanofibers inside of the porous textile substrate. The printed serpentine interconnects had a conductivity of ≈3246 S cm^{-1}. The reliability tests revealed that the increase in the resistance ratio value was approximately five times over 1000 cycles of 30% uniaxial stretching. Controlling the thickness of the cladded-layer enabled the fabrication of two-layered e-textiles, which provided a remarkable advantage in designing electrodes for sEMG and EEG applications. With our custom-designed data acquisition/transmission electronics, a complete wireless sEMG system was successfully demonstrated. We anticipate that the design principles of our two-layered e-textiles will inspire new ways of engineering innovative, practical wearable health care systems.

4. Experimental Section

Substrate Material and Processing: The textile substrate is NANOSAN-Sorb (Nanofiber Technology, Limited Liability Company), a nonwoven

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textile of thermoplastic polyurethane nanofibers. The textile has a moderate absorbency from 2.5 to 18.5 g g^{-1} (i.e., absorbed liquid/textile in mass) when soaked in saline solution (0.9% NaCl in water) for 30 min, according to nonwovens standard procedures (NWSP) 240.0.R2 (15) standard test. In addition, its flexibility, elasticity, resilience, and ductility make the textile suitable for biomedical applications as it is both sweat and blood absorbent. Therefore, the textile was employed as the substrate for the sEMG and EEG applications. Prior to printing, inner and outer surfaces of the porous substrate were treated with oxygen plasma at 1 mTorr vacuum condition (PDC-001, Harrick Plasma) for 10 min.

Nanocomposite Ink Preparation: Flouroelastomer (DAI-EL G801, Daikin Industries) and MEK (M209-1, Sigma-Aldrich) were mixed at the weight ratio of 1:2.55 with a magnetic stirrer for 6 h. After the mixing, two types of silver particles, namely Ag-flake (Sigma-Aldrich; nominal average flake size of 10 µm) and Ag-powder (Sigma-Aldrich; nominal average particle size of 2–3.5 µm) were added into the respective solution at the ratio of 2.55:1 (weight ratio between silver particle and electromeric flouropolymer), followed by mixing with magnetic stirrer for 4 h. Subsequent bath sonication for 10 min ensured a well-dispersed ink for printing. All procedures are carried out at the room temperature.

Printing of Nanocomposite Ink: The prepared ink was loaded to a custom-designed syringe (Tabletop-3Dn, nScrypt; Figure S1, Supporting Information). A sheet of the PU nanotextile (NANOSAN-Sorb) was cut into a rectangular shape of 120 mm × 120 mm and pretreated in oxygen plasma for 10 min. The nanotextile sheet was placed in the printer bed which was kept at room temperature during printing. For each printing, 10 mL of the prepared ink was loaded to a syringe. After the printing, air drying at room temperature was sufficient for all applications. The samples used for analysis were dried in vacuum chamber at room temperature for 8 h for complete removal of the solvent.

Two-Layered e-Textile Patches: After printing one side, the samples were flipped to the other side and loaded to the printer. The samples were carefully aligned with the aid of laser marking. After printing patterns on both sides, VIA were formed by puncturing the e-textile using a 3/8" diameter needle, which was dip coated in the prepared ink.

EEG Recording: For the EEG experiments, data were collected using a Brain Products V-amp amplifier. Commercial gel electrodes were used as both the reference and the ground channel, while the comparison was made between commercial and e-textile electrodes placed behind either ear. The commercial electrode was placed on the left mastoid, and the e-textile was placed on the right. Medical tape was used to electrodes in place against the skin as needed. Data were recorded at 20 000 Hz with a resolution of 24 bits (0.049 µV steps), with a 0.1 Hz high pass filter to remove drifts, and a 30 Hz low pass filter to remove high frequency noise as well as a notch filter at 60 Hz to further eliminate electrical line noise. Electrode connections to the skin were adjusted until sufficient signal quality was observed in the data. Offline, data were down-sampled to 1000 Hz, and the mean was subtracted from each channel. A wavelet analysis was employed to estimate the frequency domain signal, using frequencies between 1 and 30 Hz in 0.5 Hz steps, and a wavenumber of 20 (in order to focus on longer term changes in brain activity). The mean of this spectrogram over time was compared in each condition in order to estimate the spectra separately for eyes open and eyes closed.

The participants were seated in a dimly lit faraday chamber and were instructed to keep their eyes open and stare at a fixation screen for 3 min. After a tone was played, the participant then closed their eyes for 7 min, followed by another 2 min of staring with eyes open at the fixation. The participant did not move, kept their head still, and refrained from blinking as best possible. All procedures and protocols were approved by the local human ethics research board and the participant gave informed consent before participating.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

J.R. is the founder and CEO of True Angle Medical Technologies, a company that has commercial interests in skin-mounted electronics. All the other authors declare no conflict of interest.

Keywords

e-textiles, stretchable electronics, wetting, surface electromyography, electroencephalography

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